

Shielded Settle

Recover the real mtIndex. Make contract-held shielded coins spendable again.

Individual submission · Compact escrow + TypeScript kit + settlement desk

firstFree returns 0

A Compact contract can receive a shielded coin and still have no documented way to read that coin's Merkle index. The next spend fails.

```
invalid index into sparse merkle tree: 0
```

What builders do

Pass `queryZSwapAndContractState(...).firstFree` into `sendShielded` as `mtIndex`.

What breaks

For contract-owned coins, `firstFree` often lies. Escrow, marketplace, and treasury payouts stall after deposit.

Evidence: [midnightntwrk/servicedesk#187](#)

[Midnight forum thread on contract-owned mtIndex](#)

Documented query vs kit

DOCUMENTED QUERY

0

firstFree · proof rejects index 0

SHIELDED SETTLE

46 · 51 · 80...

real mtIndex · release or refund spends

- 01 Deserialize zswap chain state from the indexer
- 02 Call debug toString(true)
- 03 Locate the commitment owned by the contract
- 04 Qualify with resolveContractCoinMtIndex

Temporary workaround until a first-class API exists

Keep the kit call shape when #187 lands

Settlement desk in five clicks

- 01 **Deploy** — create the escrow vault on the simulated ledger
- 02 **Deposit** — `receiveShielded`; coin enters the tree
- 03 **Probe** — documented `firstFree` (expect 0)
- 04 **Resolve** — kit recovers the real Merkle position
- 05 **Release / Refund** — spend once; second spend is blocked

What Compact proves & hides

On ledger

Depositor / beneficiary identities, funded and settled flags, vault occupancy.

Shielded

Coin value and commitment stay private. Spend needs a correct
`QualifiedShieldedCoinInfo`.

Why privacy

Escrow should not force amounts or payment history into the clear.

Circuits: `deposit` → `receiveShielded` · `release / refund`
→ `sendShielded`

`contracts/escrow.compact`

Witness must carry the real `mtIndex` across transactions

One call to qualify the coin

```
const result = resolveContractCoinMtIndex({
  snapshot: { firstFree: 0n, debugDump },
  lookup: { contractAddress, coin },
});

if (!result.ok || !result.qualified) throw new Error(result.detail);
await release(result.qualified);
```

Boundary

The kit qualifies. Lace or an HTTP proof server still builds the proof.

Safety

Double-spend blocked · wrong-contract dump refused · party collision refused · 31 fund-safety tests.

What judges can open

Desk

Operator console. Signal band: coral lie · ice ledger · mint kit.

Stats

Live session KPIs and charts from deals settled in this tab.

Gap · Integrate · Docs · Demo

Evidence for #187, copy-ready snippet, run path, and video walkthrough.

One job: make contract-held shielded coins spendable again.

Shielded Settle · Midnight Korea Hackathon 2026

github.com/panagot/Shielded-Settle